

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5

Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I **MADHU BAVANA.B(192524103)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MADHU BAVANA.B

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

1. Represent the Statements in Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground becomes wet.

The given statements are:

1. If it rains, then the ground becomes wet:
 $R \rightarrow W$
2. It is raining:
 R

Goal: Prove **W**.

2. Convert All Statements into CNF

The implication is converted as:

$$R \rightarrow W \equiv \neg R \vee W$$

Therefore, the CNF clauses are:

- **C1:** $\neg R \vee W$
- **C2:** R

3. Negate the Goal

Goal = **W**

Negated goal:

$$\neg W$$

Add it as:

- **C3:** $\neg W$

4. Apply the Resolution Algorithm Step by Step

Resolve **C1** and **C2**:

$$C1: \neg R \vee W$$

$$C2: R$$

C4: W

Now resolve **C4** and **C3**:

C4: W

C3: $\neg W$

C5: $\square$

5. Empty Clause and Conclusion

The **Empty Clause** ($\square$) is obtained. Therefore, the negated goal $\neg W$ is false.

Conclusion: Hence, the ground is wet is proved using the Resolution Algorithm.

Question 2 – Student Assignment Submission

1. Write the Propositional Logic Expressions

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

The given statements are:

1. If Rahul submits the assignment, then Rahul receives internal marks:
 $S \rightarrow M$
2. Rahul submitted the assignment:
 S

Goal: Prove **M**.

2. Convert the Premises into CNF

The implication:

$S \rightarrow M$

is converted into:

$\neg S \vee M$

Therefore:

- **C1:** $\neg S \vee M$

- C2: S

3. Negate the Goal

Goal = M

Negated goal:

$\neg M$

Add:

- C3: $\neg M$

4. Apply the Resolution Algorithm

Resolve C1 and C2:

```
C1:  $\neg S \vee M$ 
C2: S
-----
C4: M
```

Now resolve C4 and C3:

```
C4: M
C3:  $\neg M$ 
-----
C5:  $\square$ 
```

5. Conclusion

The **Empty Clause** ($\square$) is obtained. Therefore, the negated goal $\neg M$ is contradictory.

Conclusion: Hence, **Rahul receives internal marks is proved using the Resolution Algorithm.**

Question 3 – Library Membership

1. Represent the Knowledge Base Using Propositional Logic

Let:

- L = Priya is a library member.
- B = Priya can borrow books.

The given statements are:

1. If Priya is a library member, then Priya can borrow books:
 $L \rightarrow B$
2. Priya is a library member:
 L

Goal: Prove **B**.

2. Convert Each Statement into CNF

The implication:

$$L \rightarrow B$$

is converted into:

$$\neg L \vee B$$

Therefore:

- **C1:** $\neg L \vee B$
- **C2:** L

3. Negate the Goal

Goal = **B**

Negated goal:

$$\neg B$$

Add:

- **C3:** $\neg B$

4. Perform Resolution Until No New Clauses Can Be Generated

Resolve **C1** and **C2**:

$$\begin{array}{l} \text{C1: } \neg L \vee B \\ \text{C2: } L \\ \hline \text{C4: } B \end{array}$$

Now resolve **C4** and **C3**:

$$\text{C4: } B$$

C3: $\neg B$

C5: $\square$

Since $\square$ is obtained, a contradiction is reached.

5. Final Conclusion

The Empty Clause $\square$ is derived. Therefore, the negated goal $\neg B$ is false.

Conclusion: Hence, Priya can borrow books is proved using the Resolution Algorithm.

Question 4 – Placement Eligibility

1. Convert the Statements into Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

The given statements are:

1. If Arun clears the aptitude test, then Arun is eligible for placement:
 $A \rightarrow E$
2. Arun cleared the aptitude test:
A

Goal: Prove E.

2. Write the CNF Clauses

Convert:

$A \rightarrow E$

into:

$\neg A \vee E$

Therefore:

- **C1:** $\neg A \vee E$
- **C2:** **A**

3. Negate the Goal

Goal = E

Negated goal:

$\neg E$

Add:

- **C3: $\neg E$**

4. Apply the Resolution Algorithm Step by Step

Resolve C1 and C2:

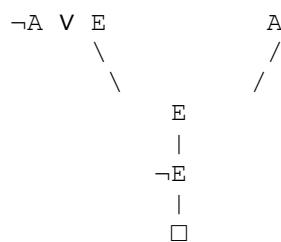
```
C1:  $\neg A \vee E$ 
C2: A
-----
C4: E
```

Now resolve C4 and C3:

```
C4: E
C3:  $\neg E$ 
-----
C5:  $\square$ 
```

5. Resolution Tree and Conclusion

Resolution Tree



Conclusion: Since the **Empty Clause** ($\square$) is obtained, the negated goal $\neg E$ is false.
Therefore, **Arun is eligible for placement is proved using the Resolution Algorithm.**

Question 5 – Access Control System

1. Represent the Knowledge Base in Propositional Logic

Let:

- **P** = User entered the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

The given statements are:

1. If the user enters the correct password, then the user is authenticated:
 $P \rightarrow A$
2. If the user is authenticated, then the user is granted access:
 $A \rightarrow G$
3. The user entered the correct password:
 P

Goal: Prove **G**.

2. Convert All Implications into CNF

First implication:

$$P \rightarrow A \equiv \neg P \vee A$$

Second implication:

$$A \rightarrow G \equiv \neg A \vee G$$

Therefore, the CNF clauses are:

- **C1:** $\neg P \vee A$
- **C2:** $\neg A \vee G$
- **C3:** P

3. Negate the Goal

Goal = **G**

Negated goal:

$$\neg G$$

Add:

- **C4:** $\neg G$

4. Apply Resolution Algorithm to Derive the Empty Clause

Step 1

Resolve C1 and C3:

```
C1:  $\neg P \vee A$   
C3:  $P$   
-----  
C5:  $A$ 
```

Step 2

Resolve C2 and C5:

```
C2:  $\neg A \vee G$   
C5:  $A$   
-----  
C6:  $G$ 
```

Step 3

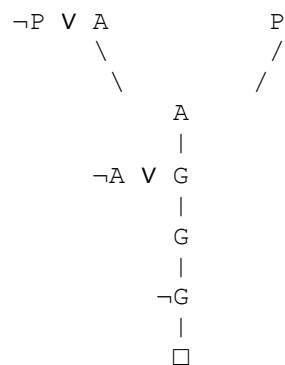
Resolve C6 and C4:

```
C6:  $G$   
C4:  $\neg G$   
-----  
C7:  $\square$ 
```

5. Explanation and Final Conclusion

The first resolution step proves that the user is **authenticated (A)** because the correct password was entered. The second resolution step proves that the authenticated user is **granted access (G)**. Finally, **G** is resolved with the negated goal $\neg G$, producing the **Empty Clause ($\square$)**.

Resolution Tree



Conclusion: Since the **Empty Clause** ($\square$) is derived, the negation of the goal leads to a contradiction. Therefore, **the user is granted access is proved using the Resolution Algorithm.**